AS5600 Connections for testing:
VCC - 3.3V
GND - GND
SDA - GPIO21
SCL - GPIO22
DIR - GND(Ensures the direction is clockwise - 0 to 360)

Sample Code:

```
#include <Wire.h>
#include <AS5600.h>

// Create an instance of the AS5600 sensor
AS5600 as5600;

// Define custom I2C pins if you want to change them (Optional)
// Default ESP32 I2C pins are GPIO 21 (SDA) and GPIO 22 (SCL)
#define I2C_SDA 21
#define I2C_SCL 22

void setup() {
  // Initialize Serial Monitor
  Serial.begin(115200);
  delay(1000);
  Serial.println("--- AS5600 ESP32 Reading Setup ---");

  // Initialize I2C bus with specific pins
  Wire.begin(I2C_SDA, I2C_SCL);

  // Connect to the AS5600 sensor
  as5600.begin();

  // Verify connection
  if (as5600.isConnected()) {
    Serial.println("AS5600 sensor detected successfully!");
  } else {
    Serial.println("Error: AS5600 not found. Check your wiring and I2C
pull-up resistors.");
    while (1); // Halt execution if sensor is missing
  }
}

void main_loop() {
  // 1. Read raw 12-bit position (0 to 4095)
  uint16_t raw_pos = as5600.readAngle();
```

```
  // 2. Convert position to degrees (0.0 to 360.0)
  float degrees = raw_pos * 360.0 / 4095.0;

  // Print results to Serial Monitor
  Serial.print("Raw Value: ");
  Serial.print(raw_pos);
  Serial.print(" \t| Angle: ");
  Serial.print(degrees, 2); // Print with 2 decimal places
  Serial.println("°");

  delay(100); // Sample rate limit (100ms)
}

void loop() {
  main_loop();
}
```

Results:
The sensor is giving perfect values from 0 to 4095(Because 12bit) which is 0 to 360 degrees

To-do: Connect the sensor in the setup, and make the motor work and take readings, then connect all this to 7semi and take readings in 1mm increment and make a table.

Connections for the setup:

The connections for R100(ESC speed controller) and AS5600 are the exact same.
R100's IN- is connected to IBT_2's B+
R100's IN+ is connected to +ve terminal of external dc power supply
-ve terminal of power supply is connected to B- of IBT_2 and B- is then connected to GND of esp32(to ensure the IBT_2 is connected to Esp32 somehow)
Brown wire of the motor is connected to M+ of IBT_2 and black to M- of IBT_2

IBT_2 Pin	ESP32 Pin	Description
VCC	5V / Vin	Powers the IBT_2 logic chips
GND	GND	Must share a common ground with the ESP32
RPWM	GPIO 25	Right/Forward PWM control
LPWM	GPIO 26	Left/Reverse PWM control
R_EN	5V / Vin (or a GPIO)	Enable Right side (bridge to 5V to keep permanently active)
L_EN	5V / Vin (or a GPIO)	Enable Left side (bridge to 5V to keep permanently active)
R_IS	*Leave Open*	Current sense output (not needed for basic control)
L_IS	*Leave Open*	Current sense output (not needed for basic control)

Code for the full setup:

```
#include <Wire.h>
#include <AS5600.h>
#include <INA226.h>

// --- Pin Definitions ---
#define I2C_SDA     21
#define I2C_SCL     22
#define IBT2_RPWM   25
#define IBT2_LPWM   26

// --- New Enable Pin Definitions (Option 2) ---
#define IBT2_REN    18  // Connected to R_EN on IBT_2
#define IBT2_LEN    19  // Connected to L_EN on IBT_2

// --- PWM Hardware Settings for ESP32 ---
#define PWM_FREQ    5000
#define PWM_RES     8  // 8-bit resolution (values from 0 to 255)
#define CH_FORWARD  0
#define CH_REVERSE  1

// --- Sensor Objects ---
AS5600 as5600;
INA226 ina226(0x40); // 0x40 is the standard default I2C address for
INA226

unsigned long lastUpdate = 0;
const int updateInterval = 250; // Read sensors every 250ms

void setup() {
  Serial.begin(115200);
  delay(1000);
  Serial.println("\n--- ESP32 Control Dashboard Initializing ---");

  // 1. Configure and Activating Driver Enable Pins
  pinMode(IBT2_REN, OUTPUT);
  pinMode(IBT2_LEN, OUTPUT);
  digitalWrite(IBT2_REN, HIGH); // Turn on Right side of H-Bridge
  digitalWrite(IBT2_LEN, HIGH); // Turn on Left side of H-Bridge
  Serial.println("[OK] IBT_2 Enable pins set HIGH.");
```

```
  // 2. Initialize I2C Bus (This configures the pins for BOTH sensors)
  Wire.begin(I2C_SDA, I2C_SCL);

  // 3. Initialize AS5600 Encoder
  as5600.begin();
  if (as5600.isConnected()) {
    Serial.println("[OK] AS5600 Encoder connected.");
  } else {
    Serial.println("[ERROR] AS5600 Encoder not found! Check I2C
wiring.");
  }

  // 4. Initialize INA226 Power Monitor (Corrected Line)
  if (ina226.begin()) {
    ina226.setMaxCurrentShunt(10.0, 0.1); // 0.1 Ohm for R100 shunt
resistor
    Serial.println("[OK] INA226 Current Monitor connected.");
  } else {
    Serial.println("[ERROR] INA226 not found! Check I2C address and
wiring.");
  }

  // 5. Configure ESP32 PWM Control Channels for IBT_2
  ledcAttachChannel(IBT2_RPWM, PWM_FREQ, PWM_RES, CH_FORWARD);
  ledcAttachChannel(IBT2_LPWM, PWM_FREQ, PWM_RES, CH_REVERSE);

  Serial.println("System Ready. Beginning Motor Sweep Loop...\n");
  delay(2000);
}


void loop() {
  // --- STAGE 1: Move Motor Forward ---
  Serial.println(">>> Action: Driving Forward...");
  ledcWriteChannel(CH_REVERSE, 0);       // Safety: Turn off opposite
direction first
  ledcWriteChannel(CH_FORWARD, 120);     // Set speed (~47% duty cycle)

  // Run for 3 seconds while monitoring status
  unsigned long stageStart = millis();
  while (millis() - stageStart < 3000) {
    runDashboard();
  }
```

```
  // --- STAGE 2: Stop and Rest ---
  Serial.println(">>> Action: Braking / Coasting...");
  ledcWriteChannel(CH_FORWARD, 0);
  ledcWriteChannel(CH_REVERSE, 0);

  stageStart = millis();
  while (millis() - stageStart < 1500) {
    runDashboard();
  }

  // --- STAGE 3: Move Motor Backward ---
  Serial.println(">>> Action: Driving Backward...");
  ledcWriteChannel(CH_FORWARD, 0);       // Safety: Turn off opposite
direction first
  ledcWriteChannel(CH_REVERSE, 120);     // Set speed (~47% duty cycle)

  stageStart = millis();
  while (millis() - stageStart < 3000) {
    runDashboard();
  }

  // --- STAGE 4: Stop and Rest ---
  Serial.println(">>> Action: Braking / Coasting...");
  ledcWriteChannel(CH_FORWARD, 0);
  ledcWriteChannel(CH_REVERSE, 0);

  stageStart = millis();
  while (millis() - stageStart < 1500) {
    runDashboard();
  }
}

void runDashboard() {
  if (millis() - lastUpdate >= updateInterval) {
    lastUpdate = millis();

    // Read Encoder Position
    uint16_t raw_pos = as5600.readAngle();
    float angle_deg = raw_pos * 360.0 / 4095.0;

    // Read Power Specs from INA226
    float voltage = ina226.getBusVoltage();
```

```
    float current_mA = ina226.getCurrent_mA();
    float current_amps = current_mA / 1000.0;

    // Print to Serial Monitor in a clean telemetry layout
    Serial.print("POS: ");
    Serial.print(angle_deg, 1);
    Serial.print("° \t| VOLT: ");
    Serial.print(voltage, 2);
    Serial.print("V \t| CURR: ");
    Serial.print(current_amps, 3);
    Serial.println(" A");
  }
}
```

This makes the motor sweep from 0 to 360 and then backwards which was successfully done.

Results:

```
POS: 302.2°     | VOLT: 11.99V  | CURR: 0.000 A
POS: 94.7°      | VOLT: 11.99V  | CURR: 0.000 A
POS: 248.9°     | VOLT: 11.98V  | CURR: 0.000 A
POS: 42.4°      | VOLT: 11.98V  | CURR: 0.000 A
POS: 195.3°     | VOLT: 11.98V  | CURR: 0.000 A
POS: 348.3°     | VOLT: 11.99V  | CURR: 0.000 A
POS: 141.7°     | VOLT: 11.98V  | CURR: 0.000 A
POS: 293.9°     | VOLT: 11.9
```